

Which alignment targets are more robust to ‘persona’ drift? measured through drift in value rankings and persona vector activations

Background / Related work

- 1) Li et al. (<https://arxiv.org/pdf/2402.10962Z>) show that system prompting is not enough to control instruction drifting over multi-turn dialogs
- 2) Geng et al. (<https://arxiv.org/pdf/2511.01805>) show that context accumulation changes LLMs beliefs
- 3) <https://www.anthropic.com/research/persona-vectors> introduce personality vectors, show how to extract them, how to monitor during deployment, and why they are important
- 4) <https://www.anthropic.com/research/assistant-axis> argue that during pre-training the model learns to simulate many characters (heros, villains, etc..), during post-training we highlight the “assistant” character, and that “assistant-like behavior” is associated with activations in a particular “assistant axis”. They also show assistant-axis drifting over multi-turn conversation in specific domains, which are domains we may want to pay closer attention to.

I propose:

- 1) Swap “instruction drifting” with “persona drifting”
 - a) Value ranking drifting
 - b) Personality vector drifting
 - c) Extra: Assistant axis drifting
- 2) Keep multi-turn dialogs, but translate to revolve around particular values
- 3) Compare robustness across alignment targets
- 4) Try to solve persona drift:
 - a) Steer models towards values during conversation by introducing periodic value conflicts
 - b) Steer models towards “personalities” or the “assistant axis” during or before conversation to prevent persona drift. Sub-Question: which alignment targets are easiest to steer towards the “assistant axis” or personality “vectors”?
 - c) Can we modify alignment targets to induce particular personalities or resist change in personality?

Hypothesis: Models post-trained on different alignment targets will exhibit different “persona distributions”, which will be associated with drifts toward different values over multi-turn conversations. Different alignment targets and associated persona distributions will directly influence the model’s values preferences as well as their robustness to persona and value drifting. Persona drift will also vary across domains, and these domains will be different for different alignment targets.

- RLHF models will likely drift easily, and particularly towards a sycophantic personality, which will be shown in value rankings drifting towards approval-seeking values

(agreeableness) and away from accuracy. These models will show more drifting in emotional domains.

- RLAIIF (i.g. Constitution based) will show smaller magnitude of drifting overall, but may “oscillate” between competing personalities as opposed to smooth transitions during the conversation. These models will drift more in high-stakes domains (like medical advice) as opposed to emotional conversations.

Impact:

- How do values of models, trained on different alignment targets, drift over multi-turn conversations?
- How robust are different alignment targets to value drifting?
- What personas manifest from each alignment target? What are the value rankings of different personas? Are any personas more robust to persona drifting?
- Can we solve persona drift by steering towards values or personas during or before the conversation?

To Monitor drift over multi-turn conversation:

- Value ranking: similar to instructing drifting - we can elicit value rankings before and after a dialogue, feeding the conversation history in the context for each conflict. Alternatively, we can extend each individual conflict to be a multi-turn dialogue. The dialogues should ideally be related to the conflict at hand.
- Persona vector shifting: allows for more fine-grained monitoring as we can track which message in the conversation led to higher magnitudes of activation changes (Anthropic work introduced this by feeding a single prompt at a time, not dialogues)
- Extra: Assistant axis drifting: track leading component of the assistant axis (as done in the original work)

Methodology

- Take an LLM (qwen) and post-trained using different alignment targets
- Per model: run value rankings, where per conflict:
 - Original conflict is given, record model answer and personality vector activations
 - A multi turn conversation is continued, surrounding values in the conversation. Personality vector activation changes are recorded.
 - The conflict is presented again, with context of conversation history. Personality vector activations are calculated again.
- Calculate percentage of answer changes AND produce two separate value ranking (one calculated using original answers and one using post-dialogue answers)
- Calculate persona vector activation changes
- Associate alignment target <-> value preferences <-> persona activations <-> multi-turn persona drift robustness
- Steer models towards personas associated with less persona drift as a preventative measure

Timeline

- Early - mid March: Create a small set of simple multi-turn conflict scenarios and show value ranking and persona vectors drifting in base model
- Mid - Late March: Extend on ConflictScope to create a full multi-turn pipeline. Finetune base model using different alignment targets
- Early April: Run multi-turn value ranking and persona drifting. Compare early results.
- Late April: mitigate persona drifting by ... (potential methods mentioned above)
- May: Paper

Path to Impact

An understanding of which alignment targets are most robust to drift in personality over a multi-turn dialogue, as measured by value rankings and persona-vector activations. It would be ideal to also correlate alignment targets with personalities that tend to “stand their grounds” over multi turn conversations in order to steer models as a method of persona drift mitigation.

Scope

Extra: How do these personality traits correlate to likelihood of emergent misalignment?

Extra: model merging / alignment target merging / personality merging?

(<https://arxiv.org/html/2509.19727v1>)

Extra: agentic route - implicit memory (<https://arxiv.org/pdf/2602.08563>)

Extra: RLMS

Failure Modes

Failure mode	Description	Likelihood	How can/will this failure mode be mitigated? (if necessary)
Scenario generation	ConflictScope extension may be too time consuming for a short research period	med	Modifying existing datasets if exist